

HealthCare+

AI-Powered Cardiac Risk Assessment Web Application

TechSprint Hackathon 2025 – Leveraging the Power of AI
Secured 4th Position (Top 5)

Organized by Google Developer Groups On Campus – Gyan Ganga Institute of Technology & Sciences

Problem Statement

Heart disease cases are increasing rapidly among young adults in India, especially in smaller cities where regular medical checkups are often expensive and difficult to access. Early awareness and preventive guidance are critical.

HealthCare+ aims to provide early cardiac risk screening along with basic lifestyle guidance through a simple and accessible web platform.

Solution Overview

HealthCare+ uses Machine Learning to analyze **13 physiological health parameters** and predict heart health status. Based on the prediction, the system provides **personalized diet and lifestyle recommendations**.

Live Demo

Web: <https://healthcarewala.netlify.app/heart-health>

(Beta: Login with any dummy email such as test@example.com — no signup required)

GitHub Repository: <https://github.com/Ritikyadav2004/HealthcarePlus>

Key Features

- ML-based cardiac risk prediction (13 medical parameters)
- Real-time input validation using medical ranges
- Personalized diet and lifestyle dashboard
- Responsive web UI (mobile & desktop)
- FastAPI backend with /predict endpoint

- JWT-based authentication
- Model deployment ready (TensorFlow / Keras)

Project Structure

```
HealthcarePlus/
|-- frontend/           # React + Vite frontend
|-- app.py              # FastAPI backend & ML inference
|-- model.h5            # Trained TensorFlow/Keras model
|-- model.py            # Model loading & prediction logic
|-- heartdisease.csv    # Training dataset
|-- requirements.txt    # Backend dependencies
|-- README.md           # Project documentation
```

Tech Stack

Category	Technologies
Frontend	React, Vite, HTML, CSS, JavaScript, Google Fonts
Backend	Python, FastAPI
ML Model	TensorFlow, Keras
Development	Google Colab
Deployment	Netlify (Frontend)
Database	MySQL (planned for medical records)

Quick Start Guide

1. Clone the Repository

```
git clone https://github.com/Ritikyadav2004/HealthcarePlus.git
cd HealthcarePlus
```

2. Backend Setup (Python)

```
pip install -r requirements.txt
uvicorn app:app --reload --host 0.0.0.0 --port 8000
```

Swagger API Docs: <http://localhost:8000/docs>

3. Frontend Setup

```
cd frontend
npm install
npm run dev
```

Frontend URL: <http://localhost:5173>

4. Production Deployment

- **Frontend:** Netlify (already deployed)
- **Backend:** Render / Railway / VPS (Gunicorn + Uvicorn)
- **Model Serving:** FastAPI / TensorFlow Serving

5. Test Live Demo

Visit <https://healthcarewallha.netlify.app> and try with sample data.

ML Model Specifications

Input Features (13 Parameters)

1. Age (years)
2. Sex (M/F)
3. Chest Pain Type
4. Resting Blood Pressure (mmHg)
5. Serum Cholesterol (mg/dl)
6. Fasting Blood Sugar (> 120 mg/dl)
7. Resting ECG Results
8. Maximum Heart Rate (bpm)
9. Exercise Induced Angina
10. ST Slope
11. Number of Major Vessels
12. Thalassemia
13. Slope of Peak Exercise ST

Output:

- Healthy
- Potential Heart Issue

The model is trained using a heart disease dataset with a TensorFlow/Keras DNN architecture.

Contributing

1. Fork the repository
2. Create a feature branch: `git checkout -b feature/YourFeature`
3. Commit your changes: `git commit -m "Add YourFeature"`
4. Push to the branch: `git push origin feature/YourFeature`
5. Open a Pull Request

License

This project is licensed under the **MIT License**.

Creator

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If you find this project useful, feel free to star the repository!